

Lecture 8: Conditioning, Bayes' Rule

Jamie Haddock

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$$f_{X|Y}(x | y) = \frac{f_{Y|X}(y | x) f_X(x)}{f_Y(y)}.$$

This holds even when one variable is discrete and the other continuous. Example: A coin has unknown bias $\Theta \sim \text{Uniform}(0, 1)$ (our *prior*). We observe k heads in n flips. The *posterior* is

$$f_{\Theta|\text{data}}(\theta) \propto \theta^k (1 - \theta)^{n-k},$$

a Beta($k + 1, n - k + 1$) density – we'll meet this machinery again in Week 15.—##
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—title: "Lecture 8: Conditioning, Bayes' Rule"author: "Jamie Haddock"format: revealjs: output-file: Lecture8_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
from scipy.stats import beta
theta_true = 0.7
n_flips = 30
flips = rng.random(n_flips) < theta_true
```

1.1 Reading & Homework
Reading: §3.6, §4.1 (Bertsekas & Tsitsiklis, Introduction to Probability).